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Characterization of diarrhoeagenic *Escherichia coli* with special reference to antimicrobial resistance isolated from hospitalized diarrhoeal patients in Kolkata (2012–2019), India

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Abstract

Aims: This study analyses the prevalence and antimicrobial resistance (AMR) of major diarrhoeagenic *Escherichia coli* (DEC) pathotypes detected in hospitalized diarrhoeal patients in Kolkata, India, during 2012–2019.

Methods and Results: A total of 8891 stool samples were collected from the Infectious Diseases Hospital, Kolkata and screened for the presence of enteric pathogens. Multiplex PCR identified the presence of DEC in 7.8% of the samples, in which ETEC was most common (47.7%) followed by EAEC (38.4%) and EPEC (13.9%). About 54% cases were due to sole DEC infections. Majority of the mixed DEC infections were caused by the *Vibrio* spp. (19.1%) followed by Rotavirus (14.1%) and *Campylobacter* spp. (8.4%). ETEC and EAEC were associated significantly with diarrhoea in children <5 years of age, whereas EPEC and also ETEC were prevalent in patients aged between 5 and 14 years. AMR profile showed high prevalence of multidrug resistance (MDR) among DEC (56.9%) in which 9% were resistant to antibiotics of six different antimicrobial classes. Screening of the AMR conferring genes of DEC showed the presence of $bla_{\text{CTX-M3}}$ (30.2%) in highest number followed by bla_{TEM} (27.5%), *tetB* (18%), *sul2* (12.6%), *strA* (11.8%), *aadA1* (9.8%), $bla_{\text{OXA-1}}$ (9%), *dfrA1* (1.6%) and bla_{SHV} (1.2%).

Conclusions: These findings highlighted the high prevalence of MDR in major DEC pathotypes that could be considered as the leading aetiological bacterial agent responsible for diarrhoea and suggests a significant public health threat.

Significance and Impact of the Study: The results of this study can help to improve the understanding of the epidemiology of DEC infections in patients with diarrhoea. Monitoring of AMR surveillance needs special attention because the DEC isolates were highly resistant to commonly used antimicrobials in the treatment of diarrhoea.

KEYWORDS

antimicrobial resistance, diarrhoeagenic *Escherichia coli*, EAEC, EPEC, ETEC

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INTRODUCTION

Diarrhoea is prevailing in the developing countries, especially in Asia and Africa, where people suffer due to poor availability of safe water and lack of proper hygiene and healthcare facilities (Kotloff et al., 2013; Mills, 2014; Troeger et al., 2018). It is the second leading cause of child death at the global level and third leading cause of child mortality in India (Fenta et al., 2020; Laxminarayan & Chaudhury, 2016). Overall, this disease kills almost 2.5 million people each year globally and 70% of them are under 5 years of age (Diarrheal disease, WHO, 2017; Fenta et al., 2020). The causative agents belong to the broad group of intestinal pathogens, which includes bacteria, viruses and parasites. Among them rotavirus, calcivirus and diarrhoeagenic *Escherichia coli* (DEC) accounted for more than half of the diarrhoeal deaths worldwide (Raghavan et al., 2017; Saka et al., 2019). A recent outbreak was also reported due to DEC in the United States where 167 people from 27 states were infected (CDC, 2019). In India, 30–40% of all diarrhoeal episodes are generally found to be associated with DEC infection (Dutta et al., 2013). Therefore, DEC has been considered as the leading diarrhoeal pathogen and needs a constant surveillance to understand the disease causing property and their antimicrobial resistance.

DEC consist of six pathotypes, among them enterotoxigenic *E. coli* (ETEC), enteroaggregative *E. coli* (EAEC) and enteropathogenic *E. coli* (EPEC) are the most common and are characterized by the presence of unique virulence factors for colonization and pathogenesis in humans (Phillips et al., 2000; Takahashi et al., 2008; Zhou et al., 2018). ETEC infections are associated with heat-stable (ST) and/or heat-labile (LT) enterotoxins encoded in the *est* and *elt* genes and fimbriae colonizing factors (CFs) (Dutta et al., 2013). In EPEC, *bfpA* and *eae* genes encode for bundle forming pilus and intimin are required for the attachment on intestinal epithelial cells (Dutta et al., 2013; Jerse et al., 1990; Rajendran et al., 2010). EAEC encode a transcriptional factor, aggregative regulator (AggR, part of the AraC family of transcription activators). AggR activates the expression of the genes *aat* and *aaiC* encoding the dispersin translocator and the chromosomal type 6 secretion system, which are present in typical EAEC (Baudry et al., 1990; Dudley et al., 2006). These virulence marker genes are unique to each pathotype and have been used in the detection of DEC (Dutta et al., 2013; Panchalingam et al., 2012).

Besides their prevalence, a major public health concern in diarrhoea endemic regions is the spread of antibiotic resistance (AMR). Antibiotics are important in treating bacterial infections and have been used for the treatment of severe diarrhoeal diseases and traveller's diarrhoea to reduce the duration of illness (Nguyen et al., 2005). Extensive use of chloramphenicol and tetracycline in the treatment for diarrhoea has increased the resistance in DEC. Hence, the use of fluoroquinolones is recommended for DEC-associated infections due to their lesser resistance (Heidary et al., 2014; Omolajaiye et al., 2020). Other *E. coli* types, such as extraintestinal pathogenic *E. coli* (ExPEC) that are related to bacteraemia and urinary tract infections, have also found to develop resistance to several antibiotics (Alhashash et al., 2013). Although antibiotics help in limiting infectious diseases, their excessive use results in the development and spread of antibiotic resistance at an alarming rate. Persistence of DEC infection is mostly due to the development of multidrug resistance (MDR) with repeated exposures of antibiotics during the course of treatment (Baym et al., 2016; Omolajaiye et al., 2020). MDR *E. coli* belongs to the critical category of pathogens, as it possesses serious threat to the patients in daily care hospitals in many countries (Rajpara et al., 2018; WHO reports, 2017). Resistance to extended spectrum β-lactamases (ESBLs), aminoglycosides, fluoroquinolones, macrolides, sulphonamides and tetracyclines are reported in clinical *E. coli* isolates (Colodner, 2005; Nepal et al., 2017; Paitan, 2018; Pons et al., 2014; Rawat & Nair, 2010). Therefore, it is very important to understand the AMR pattern so that proper medication and treatment can be provided to the patients suffering with the DEC infection (Heidary et al., 2014). The enhanced AMR is generally conferred by the acquirement of several AMR encoding genes through the mobile genetic elements (MGEs) (Natarajan et al., 2018). Hence, investigation of such AMR genes could help in understanding the role of MGEs in the spread of resistance among the DEC isolates.

Considering the importance of DEC in high rates of childhood morbidity, we conducted an 8-year study from 2012 to 2019 to investigate its prevalence in different age group of patients suffering from diarrhoea who are admitted in the Infectious Diseases and Beliaghata General Hospital (IDH) in Kolkata, India. All the DEC isolates were also tested for AMR encoding genes based on their antimicrobial resistance patterns. We have also studied the co-infection of DEC with other enteric pathogens to understand the age-specific infection and disease severity.

MATERIALS AND METHODS

Collection of clinical samples and patient history

This diarrhoeal disease surveillance study was conducted at the Infectious Diseases and Beliaghata General Hospital (IDH), in Kolkata, India. The Institutional

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Ethics Committee (IEC) of the National Institute of Cholera and Enteric Diseases (NICED) approved this study (Approval Number: A-1/2015-IEC). Informed consent was obtained from each patient or parents in the case of children and enrolled in this study. Confidentiality of all the information was maintained as per the clinical ethics. Stool specimens were collected during 2012–2019 from diarrhoeal patients. The samples were collected from every 5th diarrhoea patient who may or may not receive a treatment before admission. This fraction of patients represents a total of almost 40,000 diarrhoeal cases received treatment after hospitalization. Discharge of three or more loose stools in 24 h, with or without symptoms such as abdominal pain or cramps, vomiting, faecal urgency, dehydration or dysentery, was considered as diarrhoea. The stool samples were processed in the laboratory for the identification of enteric organisms within 2 h of collection following the standard protocols (Panchalingam et al., 2012). The clinical, demographic and laboratory data were entered in a specific data entry proforma supported with the SPSS.17.0 software (SPSS Inc, USA).

Microbial culture techniques for the isolation of DEC and other pathogens

A loopful of stool sample was streaked on MacConkey agar (Difco, BD, USA) plate for the isolation of major DEC pathotypes and incubated at 37°C for 18–20 h. Three lactose fermenting pink colonies from each sample were selected and sub-cultured on Luria agar (LA, Difco) plate. For the presumptive identification of *E. coli*, cultures from the LA plate were checked for oxidase, acid/gas production in triple sugar iron agar and indole production (Dutta et al., 2013; Panchalingam et al., 2012). Further confirmation was performed using multiplex PCR for the detection of virulence marker genes for each pathotype. Other enteric pathogens were isolated and identified following the published procedures (Panchalingam et al., 2012).

Multiplex PCR assay for DEC identification

To get the DNA template for multiplex PCR, colonies were picked up from LA plate, suspended in 200 μl of phosphate-buffer saline (PBS) and boiled for 10 min followed by centrifugation at 8000*g* during 10 min and supernatant containing DNA was used in PCR experiment. The multiplex PCR was carried out for the presence of specific virulence genetic markers: *est* and/or *elt* for ETEC, *eae* and/or *bfpA* for EPEC, and *aatA* and/or *aaiC* for EAEC. A total of six primer pairs were used to detect the different virulence-associated genes (Table S1). PCR was performed following published procedures (Panchalingam et al., 2012). Presence of at least one of the virulence genes was considered as positive for the respective DEC pathotype (Dutta et al., 2013; Panchalingam et al., 2012).

Antimicrobial susceptibility testing

A total of 255 DEC isolates were selected for antimicrobial susceptibility testing. These isolates have covered each DEC pathotypes (107 EAEC, 47 EPEC and 101 ETEC) and represented almost one-third of the isolates from each year. Kirby-Bauer disc diffusion method was followed using 12 different antimicrobials, including ceftazidime (30 μg), ampicillin (10 μg), ceftriaxone (30 μg), meropenem (10 μg), gentamicin (10 μg), sulphamethoxazole–trimethoprim (23.75 and 1.25 μg, respectively), streptomycin (10 μg), tetracycline (30 μg), doxycycline (30 μg), ciprofloxacin (5 μg), ofloxacin (5 μg) and chloramphenicol (30 μg). Fresh cultures were grown at 37°C in Luria-Berteni (LB) broth until they reach 0.5 McFarland optical density. The culture was then spread onto Muller-Hinton agar (MHA, Difco) plates before placing the antimicrobial discs (BD, BBL, USA). Zone diameters were measured after an incubation of 18–20 h at 37°C as per the Clinical and Laboratory Standards Institute guideline (CLSI, 2019). The *E. coli* strain ATCC 25922 was taken as a control for checking the disc accuracy (CLSI, 2019).

PCR for AMR encoding genes

MDR strains were subjected to PCR for the presence of the following AMR determining genes: bla_{TEM}, $bla_{OXA\text{-}1}$, $bla_{CTX\text{-}M3}$, bla_{SHV} for β-lactams (Maynard et al., 2003); *qnrB*, *qnrS* and *aac6′-Ib-cr* for plasmid-mediated quinolone resistance, *sul2* and *dfrA1* for sulfamethoxazole–trimethoprim (SXT) resistance; *strA* and *aadA* for streptomycin (aminoglycoside); *catI* for chloramphenicol and *tetB* for tetracycline group of antibiotics using the primer sets listed in Table S1 (Marbou et al., 2020; Sarkar et al., 2015; Sunde & Norström, 2005). The PCR mixture (total volume, 30 μl) consists of a final concentration of 1X Standard Taq Reaction Buffer (NEB, USA), 0.33 mM dNTP mixture, 0.33 μM of each forward and reverse primer (Sigma), 1 U of Taq DNA Polymerase (NEB, USA) and 50–80 ng of template DNA. Amplicon size and the annealing temperatures of the primers for each target gene are furnished in Table S1.

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Statistical analysis

The patient's age was classified into six age groups comprising of ≤2, >2–5, >5–14, >14–30, >30–50 and ≥50 years (Dutta et al., 2013). The DEC prevalence was aligned with the age groups along with the pathogens. Multinomial logistic regression analysis was performed. The differences marked as statistically significant that yield *p*-values of <0.05. Odds ratio (OR) along with the 95% confidence interval (CI) was also calculated to determine the age group at significant risk.

RESULTS

Detection of DEC and other pathogens

During an 8-year diarrhoeal disease surveillance period from 2012 to 2019, 8891 patients admitted in the IDH were enrolled for this study. Based on the selection strategy, this number represents almost one-fifth of the total patients who received treatment in the hospital. DEC was identified in 7.8% (690/8891) of the patients. Among DEC, ETEC and EAEC remained predominant pathotypes (47.7% and 38.4%, respectively) (Table 1). Among the ETEC, 55% (182/329) of the isolates harboured *elt* as well as *est*. Remaining 32% (105/329) and 13% (42/329) of the ETEC carried *est* and *elt*, respectively. In case of EPEC, 26% (25/96) of the isolates were typical carrying both the genes *bfpA* and *eae*. Among the EAEC, 38.5%, 47.2% and 14.3% of the isolates harboured *aat*, *aaiC* and *aat* and *aaiC* genes, respectively. About 54% (374/690) of the patients were identified to have DEC as a sole pathogen and 46% (316/690) of the patients had mixed pathogens, that is, the presence of the other common diarrhoea causing enteric pathogen(s) concurrently with DEC. Other enteric pathogens such as shigellae, vibrios, campylobacters, salmonellae, aeromonads, viruses and parasites were identified in the mixed infection category. *Vibrio* spp. was the most prevalent in the mixed infection category, followed by Rotavirus, *Campylobacter* spp., *Giardia* and *Shigella* spp. Other pathogens such as aeromonads, *Cryptosporidium*, *Entamoeba* and *Salmonella* spp. were rarely associated with DEC infections (Tables S1).

Seasonal isolation rate was highest during the monsoon season (June–September) for all the years followed by pre-monsoon (March–May) and post-monsoon seasons (October–December). The isolation rate was relatively low during the winter season, that is, during January and February (Indian Meteorological Department, 2022) (Figure 1).

TABLE 1 Year-wise prevalence pattern of different DEC pathotypes isolated from patients during 2012–2019

Year	No. of DEC	No. of ETEC	No. of EPEC	No. of EAEC
2012	93	37 (39.8)[a]	14 (15.0)	42 (45.2)
2013	57	25 (43.9)	8 (14.0)	24 (42.1)
2014	90	39 (43.3)	17 (18.9)	34 (37.8)
2015	152	77 (50.7)	26 (17.1)	49 (32.2)
2016	109	56 (51.4)	5 (4.6)	48 (44.0)
2017	53	37 (69.8)	4 (7.6)	12 (22.6)
2018	83	34 (41.0)	12 (14.4)	37 (44.6)
2019	53	24 (45.3)	10 (18.9)	19 (35.8)
Total (%)	690	329 (47.7)	96 (13.9)	265 (38.4)

[a]Figures in parentheses represent percentage of total against each year.

Susceptibility of different age groups for DEC infection

Sole infection frequency of DEC was predominately higher in most of the age groups than the mixed infection. The detection rate of DEC in age group ≤2 years was the highest (31.4%) followed by the age group >30–50 years (22.5%) and >14–30 years (18.6%). Further observation showed that the frequency of EAEC was higher in children ≤2 years of age (20.5%) than in adults. Similarly, the prevalence of EPEC was higher in children ≤2 years of age (5.2%) than in other age groups. Interestingly, ETEC was mostly prevalent in adults viz., >30–50 years (14.4%), >14–30 years (11.4%) and >50 years (11.5%) (Figure 2). The sole infection rate with EAEC and EPEC was observed highest in the age group of ≤2 years. However, the ETEC sole infection rate was higher in older age groups. When the mixed pathogenic infection was considered, the scenario was unchanged, that is, EAEC and EPEC showed a high

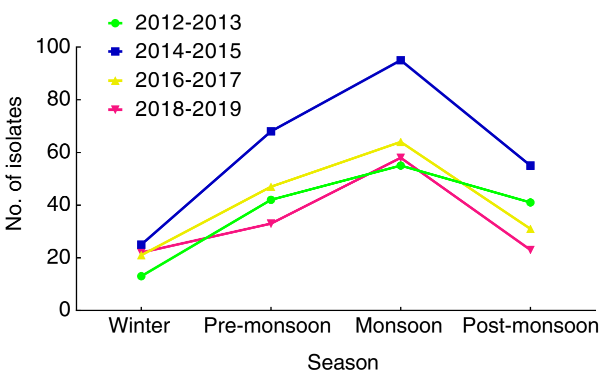

FIGURE 1 Season-wise distribution pattern of DEC isolates from the years 2012 to 2019. Monsoon season presents a higher number of DEC isolates

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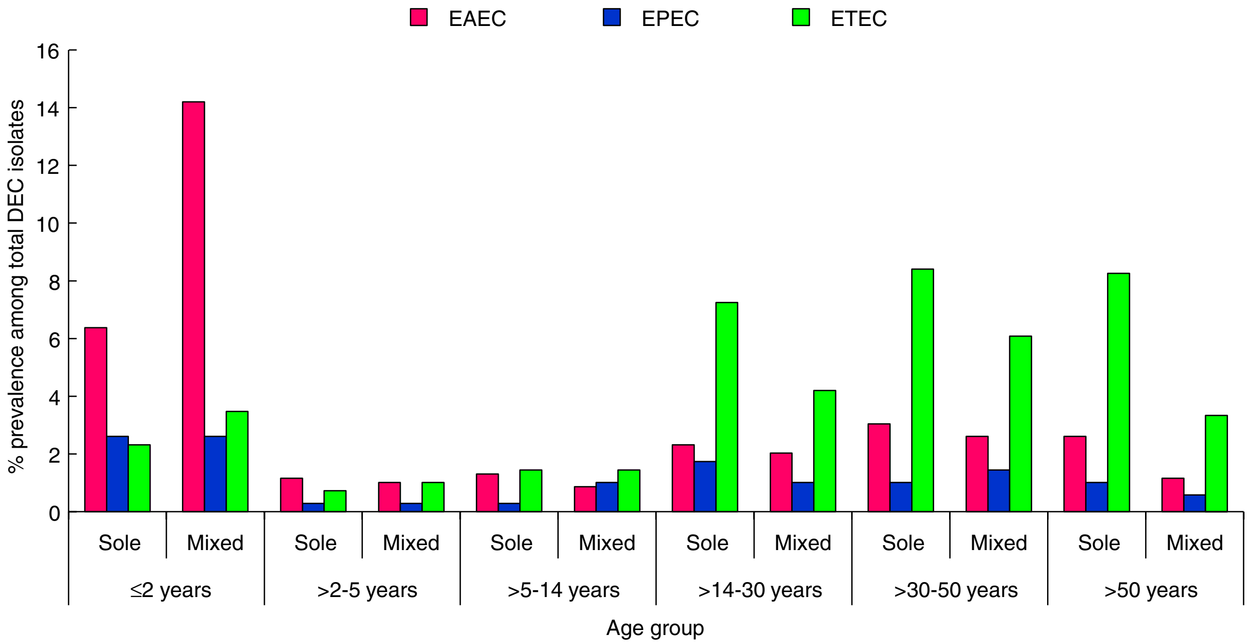

FIGURE 2 Sole and mixed pathogenic distribution of different DEC pathotypes according to the age groups of the infected patients. Figure shows high prevalence of EAEC sole and mixed infections in age group ≤2 years. ETEC shows a higher prevalence in older age groups

frequency in ≤2 than other age groups, but ETEC mixed infection was more in older age groups (Figure 2).

The most significant risk groups for DEC infection

To find out the high-risk age groups to specific DEC pathotypes, a multinomial logistic regression model was made and in this, >50 years age group was considered as the reference category. According to this model, ETEC and EAEC were significantly detected more in children ≤2 years of age ($p = 0.000$) (Table 2). It has also been found that the age group >2–5 years as a significant risk age group for ETEC and EAEC infections ($p = 0.003$ and $p = 0.005$, respectively). The age group >5–14 years is the only age group where EPEC infection was found significantly ($p = 0.049$). In this same age group, ETEC infection was also significant ($p = 0.018$) (Table 2).

Clinical symptoms of different pathotypes of DEC

Watery diarrhoea was seen in more than 60% of the DEC-infected individuals and the rate was higher (>70%) in patients who were detected with ETEC, sole EPEC and EAEC mixed with other enteric pathogens (Table 4). Bloody diarrhoea was recorded (3.4%) among EAEC-infected cases. Comparing the other clinical symptoms and disease severity caused by the different pathotypes of DEC, it was observed that vomiting is a predominant symptom (>80%) among EAEC and EPEC sole infection. Abdominal pain was mainly found among solely infected patients with ETEC and EPEC (>50%). Severe dehydration was observed in patients suffering from sole ETEC and EAEC infections (>9%). Fever was seen in >18% of the DEC-positive cases. Patients with EAEC infections suffered more and stayed about 37 h on average at the hospital for treatment (Table 3).

Antimicrobial resistance in DEC

An interesting pattern of resistance was observed for EAEC pathotype which showed resistance towards most of the antibiotics followed by EPEC and ETEC. The rate of β-lactam resistance is high in EAEC. Almost 86% of the EAEC isolates were found resistant to ampicillin, whereas 61.7% and 22.4% of the isolates showed resistance to the third-generation β-lactam antibiotics, ceftriaxone and ceftazidime, respectively. Similarly, the resistance to fluoroquinolones such as ciprofloxacin (72%) and ofloxacin (71%) has been high in EAEC. Resistance to gentamicin (15%, 8% and 5% for EAEC, EPEC and ETEC, respectively) and chloramphenicol (20%, 19% and 11% for EAEC, EPEC and ETEC, respectively) was rarely found in the DEC, but all the isolates remained susceptible to meropenem. EPEC showed highest resistance to sulphomethoxazole–trimethoprim (61.7%), tetracycline (63.8%) and doxycycline (59.6%) compared to other pathotypes. Among the ETEC, high resistance rates were observed to ampicillin

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TABLE 2 Multinomial logistic regression models exploring significant risk age group of predominant DEC infection at IDH, Kolkata

Age group in years	DEC pathotypes	B (regression coefficient)	Odds ratio (95% CI)	*p*-value
≤2	ETEC	2.275	9.72 (5.79–16.34)	0.000[a]
	EPEC	−0.63	0.53 (0.25–1.08)	0.083
	EAEC	−1.89	0.15(0.09–0.26)	0.000[a]
>2–5	ETEC	1.23	3.42 (1.50–7.78)	0.003[a]
	EPEC	−0.35	0.7 (0.20–2.37)	0.567
	EAEC	−1.2	0.3(0.13–0.69)	0.005[a]
>5–14	ETEC	0.86	2.37 (1.15–4.88)	0.018[a]
	EPEC	−0.96	0.38 (0.14–0.99)	0.049[a]
	EAEC	−0.56	0.57 (0.26–1.24)	0.157
>14–30	ETEC	0.31	1.36 (0.81–2.32)	0.243
	EPEC	−0.51	0.6 (0.27–1.32)	0.206
	EAEC	−0.105	0.9 (0.49–1.63)	0.737
>30–50	ETEC	0.15	1.16 (0.70–1.95)	0.553
	EPEC	−0.17	0.83 (0.37–1.86)	0.661
	EAEC	−0.17	0.84 (0.47–1.48)	0.554
>50	Reference category			

[a]Statistically significant.

TABLE 3 Clinical attributes of patients identified with sole and mixed infections with different pathotypes of DEC

	EAEC		EPEC		ETEC	
Attributes	**Sole**	**Mixed**	**Sole**	**Mixed**	**Sole**	**Mixed**
Duration of diarrhoea (before admission) (hours)	29.5 ± 1.9	34.2 ± 2.1	28.7 ± 2.9	31.7 ± 3.0	30.2 ± 1.6	29.3 ± 2.7
Duration of hospital stay (hours)	34.9 ± 1.8	41.2 ± 2.1	39.1 ± 2.7	33.8 ± 3.1	33.2 ± 1.4	34.9 ± 1.8
Dehydration status: (%)						
No	0.90	0.00	0.00	2.20	1.00	0.00
Some	89.70	95.90	91.80	91.30	89.10	91.10
Severe	9.50	4.10	8.20	6.50	9.90	8.90
Type of diarrhoea: (%)						
Watery	68.10	78.40	77.60	69.60	73.40	77.00
Loose	28.50	18.20	20.40	28.30	24.50	21.50
Blood/Mucus	3.40	3.40	2.00	2.20	2.10	1.50
Presence of clinical symptoms: (%)						
Fever	24.10	29.10	28.60	21.70	22.40	18.50
Vomiting	82.80	79.70	83.70	82.60	78.10	82.20
Abdominal pain	40.50	32.40	51.00	43.50	56.20	52.60

(71.3%), ciprofloxacin (53.5%) and ofloxacin (53.5%). Greater resistance against aminoglycosides such as streptomycin and gentamicin were noticed in EAEC followed EPEC and ETEC (Figure 3).

MDR analysis of the DEC isolates showed the presence of many patterns. About 57% (145/255) isolates were found resistant to at least three different classes of antimicrobials, while 14.9% (38/255) of the isolates were resistant to either one or two classes. Prevalence of MDR was more in EPEC (72.3%, 34/47) followed by EAEC (66.3%, 71/107) and ETEC (39.6%, 40/101). MDR isolates were grouped as MDR1, MDR2, MDR3 and MDR4 that represent the isolates resistant to at least 3, 4, 5 and 6 different classes of antimicrobials (β-lactums, sulphonamides/trimethoprims, aminoglycosides, tetracyclines, quinolones and chloramphenicols), respectively. Among the total MDR

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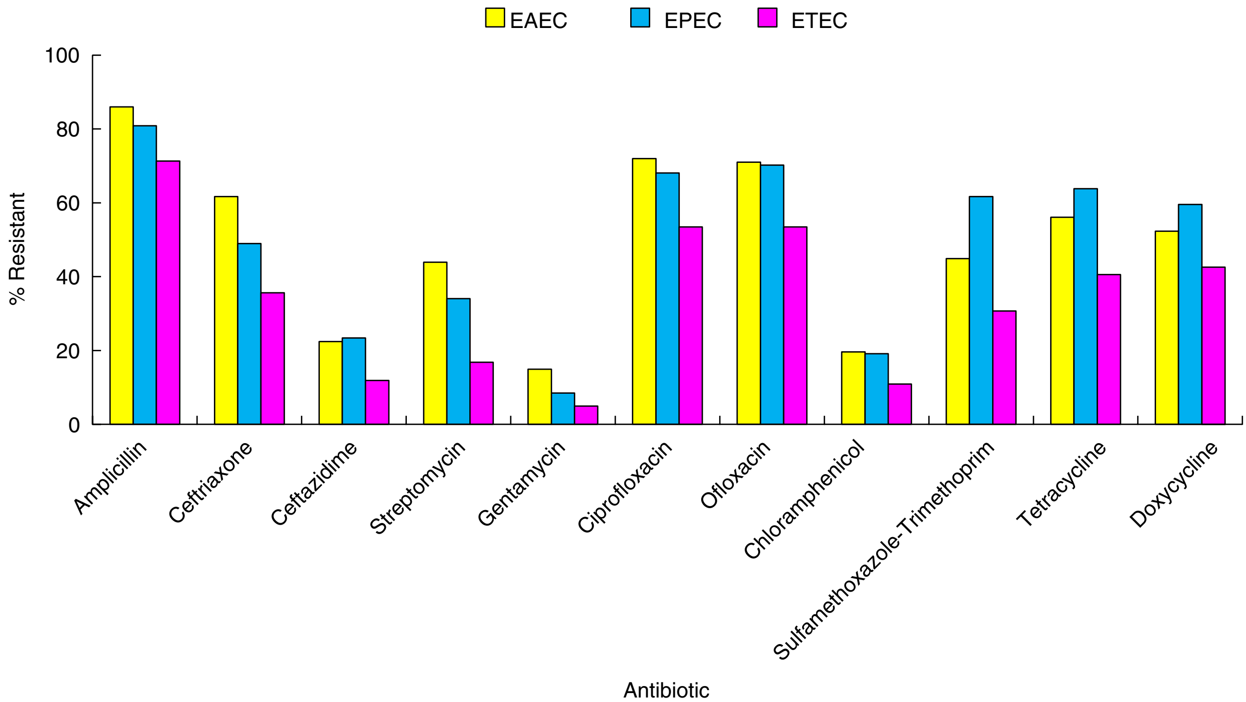

FIGURE 3 Antimicrobial resistance pattern of different DEC pathotypes isolated in Kolkata during 2012–2019

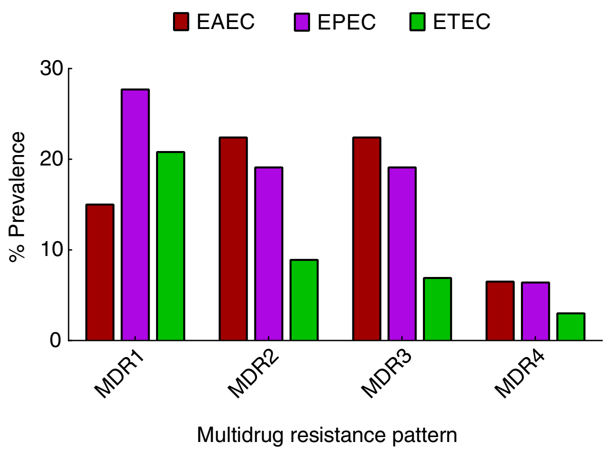

FIGURE 4 Multidrug resistance profile of different DEC isolates of Kolkata. MDR1, MDR2, MDR3 and MDR4 represent the strains those are resistant to at least 3, 4, 5 and 6 different classes of antimicrobials, respectively. Higher prevalence of MDR phenotype was found in EPEC isolates compared to other DEC pathotypes

isolates, those exhibiting the MDR1 phenotype were the most prevalent (34.5%), followed by MDR2 (29%), MDR3 (27.5%) and MDR4 (9%) (Figure 4).

Existence of different genes encoding AMR among DEC isolates

Of the 255 DEC isolates, 151 isolates that showed resistance towards β-lactam antibiotics were tested in PCR assay. Among these, 46.3% (70/151), 51% (77/151), 15.2% (23/151), 2% (3/151) isolates harboured *bla*$_{\text{TEM}}$, *bla*$_{\text{CTX-M3}}$, *bla*$_{\text{OXA-1}}$, *bla*$_{\text{SHV}}$, respectively. In 77 sulphamethoxazole–trimethoprim-resistant isolates, 41.5% (32/77) harboured *sul2* gene and 5.2% (4/77) were positive for *dfrA1*. In all, 61 isolates were resistant to streptomycin, of which 49.2% (30/61) were positive for *strA* and 41% (25/61) contained *aadA* gene. Tetracycline resistance was noticed in 104 isolates in which 44.2% (46/104) contained *tetB* gene. Quinolone resistance was shown by 131 of the isolates in which 6.9% (9/131) were positive for *qnrB* gene, 27.5% (36/131) for *qnrS* and 9.2% (12/131) for *aac6'-Ib-cr*. About 27% (11/41) of the cholramphenicol-resistant isolates harboured *cat1A* gene (Table 4). Furthermore, among the MDR isolates, *bla*$_{\text{CTX-M3}}$ and *bla*$_{\text{TEM}}$ were more prevalent, followed by *bla*$_{\text{OXA-1}}$, *sul2*, *strA*, *tetB*, *qnrS* and *qnrB*. Our analysis also showed that the presence of *bla*$_{\text{CTX-M3}}$, *bla*$_{\text{OXA-1}}$, *aadA* and *aad6'-Ib-cr* were more in EAEC. *sul2*, *dfrA1*, *strA*, *tetB*, *qnrB* and *catI* were mostly detected in EPEC, whereas *bla*$_{\text{TEM}}$, *bla*$_{\text{SHV}}$ and *qnrS* were more common among the ETEC isolates (Table 4).

DISCUSSION

DEC has emerged as one of the major public health risks in children after rotavirus and cholera (Dutta et al., 2013; Konaté et al., 2017; Takahashi et al., 2008). Due to the clinical complexity of the infection, these patients need hospitalization and administration of antimicrobials along with treatment for severe dehydration.

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TABLE 4 Prevalence of antimicrobial resistance-conferring genes among the resistant EAEC ($n = 107$), EPEC ($n = 47$) and ETEC ($n = 101$) isolates

Confers resistance to	Genes	No. of EAEC (%)	No. of EPEC (%)	No. of ETEC (%)
Beta-lactams	bla_{TEM}	28 (26.2)	12 (25.5)	30 (29.7)
	$bla_{\mathrm{OXA\text{-}1}}$	17 (15.9)	6 (12.8)	0 (0.0)
	$bla_{\mathrm{CTX\text{-}M}}$	33 (30.8)	14 (29.8)	30 (29.7)
	bla_{SHV}	1 (0.9)	0 (0.0)	2 (2.0)
Sulphonamides and trimethoprims	*sul 2*	18 (16.8)	11 (23.4)	3 (3.0)
	dfrA1	1 (0.9)	1 (2.1)	2 (2.0)
Streptomycin (Aminoglycoside)	*strA*	16 (15.0)	9 (19.1)	5 (5.0)
	aadA	22 (20.6)	1 (2.1)	2 (2.0)
Tetracyclines	*tetB*	19 (17.8)	17 (36.2)	10 (9.9)
Quinolones	*qnrB*	4 (3.7)	2 (4.3)	3 (3.0)
	qnrS	12 (11.2)	7 (14.9)	17 (16.8)
	aad6'-Ib-cr	11 (10.3)	1 (2.1)	0 (0.0)
Chloramphenicols	*catI*	3 (2.8)	5 (10.6)	3 (3.0)

Prevalence of DEC in this study is 7.8%, which is comparable to the reports from other affected countries (Lima et al., 2017; Schroeder et al., 2002; Takahashi et al., 2008). Diarrhoea caused by the DEC as sole pathogen was 54%. Mixed infections caused by DEC with other pathogens (*Vibrio*, *Campylobacter*, *Shigella*, *Salmonella*, Rotavirus and parasites) were high (46%), which is more than the previous investigations (28%) (Farfán-García et al., 2020; Nair et al., 2010). In addition, several aspects of present observation differed from a previous study conducted in Kolkata (Dutta et al., 2013), which includes (i) higher incident rate (54%) of DEC sole infection, (ii) ETEC was found to be mostly associated with diarrhoea as against EAEC and (iii) infections caused by ETEC containing both the ST and LT toxins were higher compared to the other ETEC with LT or ST alone. Although EAEC and EPEC are more associated with diarrhoea in countries like Brazil, Tanzania and several parts of India including Northern India and Karnataka (Canizalez-Roman et al., 2013, 2016; Dutta et al., 2013; Okeke et al., 2000; Shetty et al., 2012; Singh et al., 2019; Takahashi et al., 2008), our study shows that ETEC as the major DEC pathotype.

DEC-associated diarrhoea was found to vary among different age groups of the patients. Similar to the previous finding, ETEC and EAEC infections were significantly higher in younger age groups (Dutta et al., 2013; Rajendran et al., 2010). However, compared to the other reports from Vietnam and India, ETEC was higher in age group >30–50 years (Natarajan et al., 2018; Nguyen et al., 2006). Therefore, it is important to consider older age groups for monitoring DEC infections.

Similar to our previous studies, DEC-associated clinical symptoms are much less compared to cholera and Rotavirus gastroenteritis (Dutta et al., 2013; Nair et al., 2010). A recent study in Cameroon suggested that among DEC-associated diarrhoeal patients, only fever and vomiting were seen in EPEC-infected cases (Marbou et al., 2020). However, in Kolkata, these symptoms remain prominent in all the DEC cases regardless of their pathotypes. Globally, the seasonal prevalence of DEC was more during summer and rainy seasons (Ahmed et al., 2013; Behiry et al., 2011; Estrada-Garcia et al., 2009; Zhou et al., 2018). We have seen a higher incidence of DEC during 2015, and this might be associated with heavy Southwest monsoon and floods in several Districts of West Bengal, contaminating the water resources.

Widespread and arbitrary use of antimicrobials has increased the resistance ability of many bacteria, which also leads to the emergence of highly pathogenic MDR strains. In this study, majority of the DEC isolates have been identified as MDR, which was relatively higher than that of the other reports (Abbasi et al., 2020; Ghorbani-Dalini et al., 2015; Ramirez Castillo et al., 2013; Shah et al., 2015; Zhou et al., 2018). Furthermore, our findings showed a high prevalence of fluoroquinolone resistance in DEC isolates, which is one of the drugs advocated for the treatment of diarrhoea (Alikhani et al., 2013; Heidary et al., 2014; Omolajaiye et al., 2020). Studies conducted in China and Mexico also have shown higher resistance of DEC to ampicillin, sulfamethoxazole–trimethoprim and tetracycline (Canizalez-Roman et al., 2016; Zhou et al., 2018; Zhu et al., 2016). Most of the DEC isolates showed resistance to fluoroquinolone drugs demonstrating resistance may be acquired by horizontal gene transfer similar to the observation made by Ciesielczuk et al. (2013). Similar to a study conducted in France, we found that DEC from Kolkata has also

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shown β-lactam resistance mostly with isolates harbouring *bla*$_{CTX-M3}$, *bla*$_{TEM}$, *bla*$_{OXA-1}$ and *bla*$_{SHV}$ (Blanc et al., 2014). The presence of several plasmid-mediated quinolone resistance genes such as *qnrS*, *qnrB* and *aac6′-Ib-cr* has been encountered in this study. This trend has been reported in other investigations also (Herrera-León et al., 2016; Pazhani et al., 2011). However, the prevalence of *aadA*, *catA1* and *tetB* gene was relatively low as reported from Iran and South Africa (Heidary et al., 2014; Omolajaiye et al., 2020). EPEC isolates were found to harbour most of the AMR conferring genes followed by EAEC and ETEC.

This study highlights the important contribution of DEC in the prevalence of acute diarrhoea and also reveals existence of MDR among several pathotypes challenging the success of clinical management of diarrhoea.

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CONFLICT OF INTEREST

No conflict of interest declared.

AUTHORS' CONTRIBUTION

DG, PS, TR and AKM conceptualized the study. DG, GC, PS and SS performed the experiments. DG, PS and AKM prepared the original draft of the manuscript. AD performed the statistical analysis. MB, AM, SM, SD and AKM collaborated in this study. All the authors did data analysis, draft review, editing and approval.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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